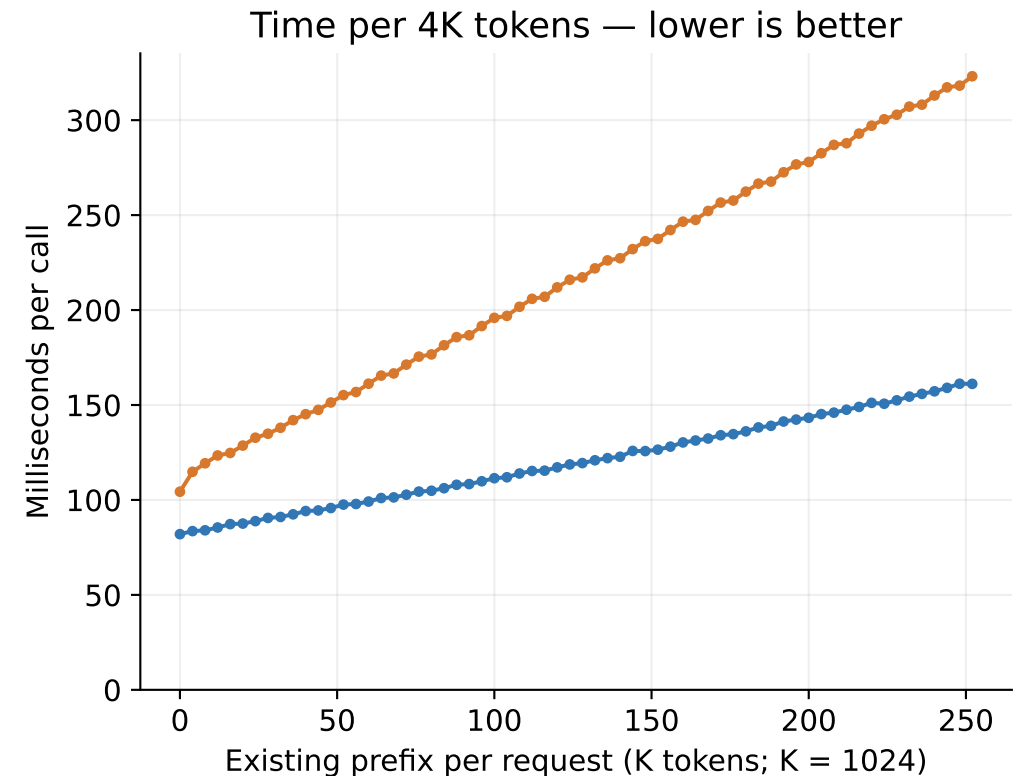
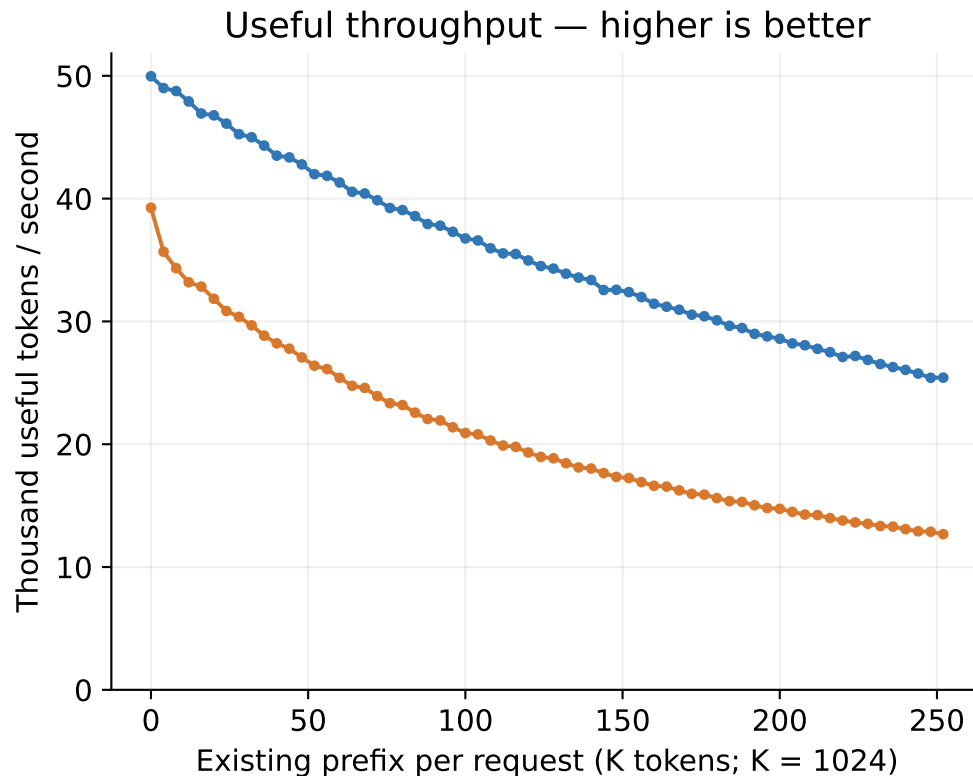


Fixed 4×1K batching vs canonical 4K

Same token count per call: 4,096 useful tokens. One request × 4K versus four requests × 1K.

— Canonical: 1 request × 4K — Fixed batch: 4 requests × 1K



0K prefix: canonical 82.0 ms; fixed batch 104.3 ms. The batch takes 27% longer for the same token count.

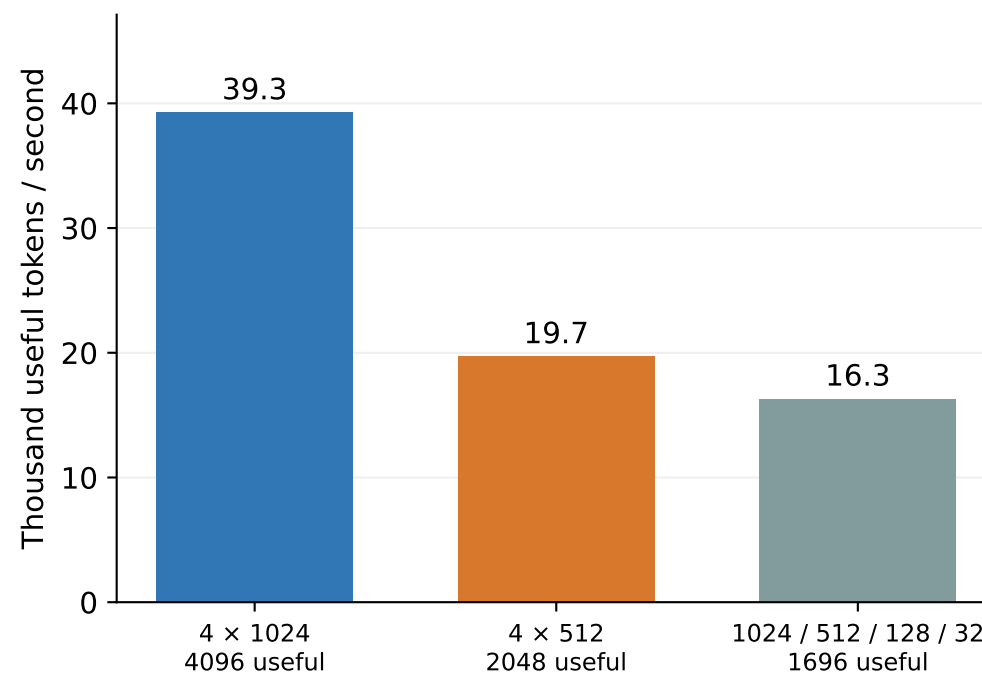
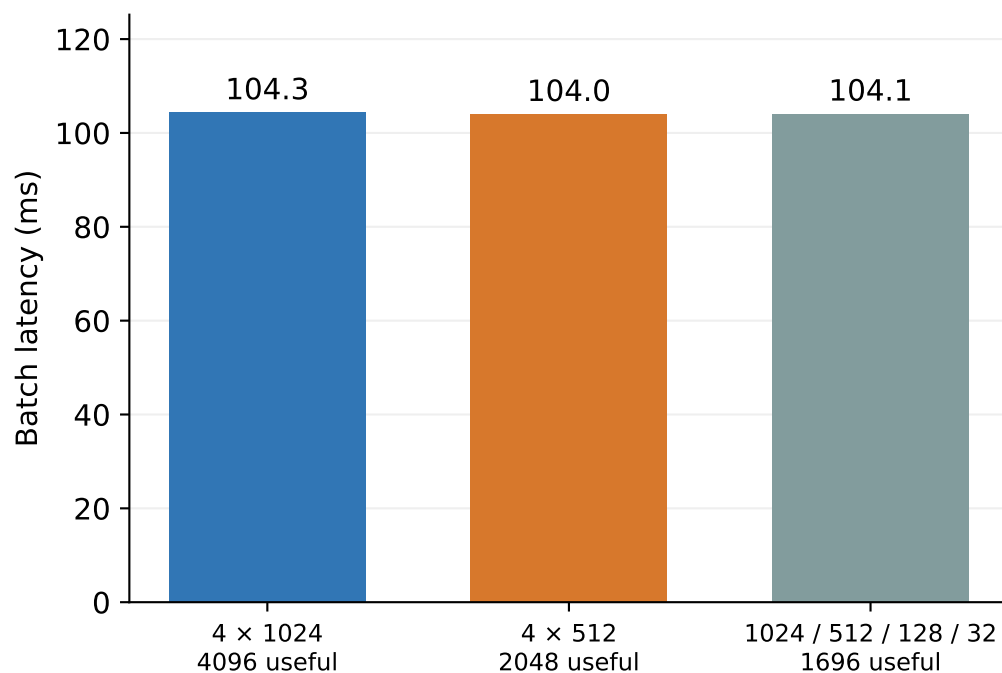
128K prefix: canonical 119.4 ms; fixed batch 217.2 ms. The batch takes 82% longer for the same token count.

248K prefix: canonical 161.2 ms; fixed batch 318.2 ms. The batch takes 97% longer for the same token count.

Gemma-4-31B-it, 60 layers, Blackhole CP8/TP4, original reduce-scatter, default activation placement. Each call processes 4,096 useful tokens: one 4K request in canonical, or four 1K requests in the fixed batch. Every request has the prefix shown on the chart. All histories were populated by real model calls. Five replays at prefixes 0, 8K, 32K, 64K, 128K, 192K and 248K; one replay at other positions. Timing covers trace replay and synchronization. Host staging is recorded separately; loading, compilation, output download and KV migration are excluded.

What does fixed-size padding cost?

All cases execute the same $4 \times 1K$ shape. Short final requests still occupy their full 1K lane.



Why measure this? A fixed batch always computes 1K rows per request when fewer tokens are useful. All three cases start at zero and reuse the same trace.

The model shares embedding, projections, norms, RoPE and MLP across four requests. Each attention layer runs four sequential 1K attention calls with local slices and one local concatenation. It adds no CP all-gathers.

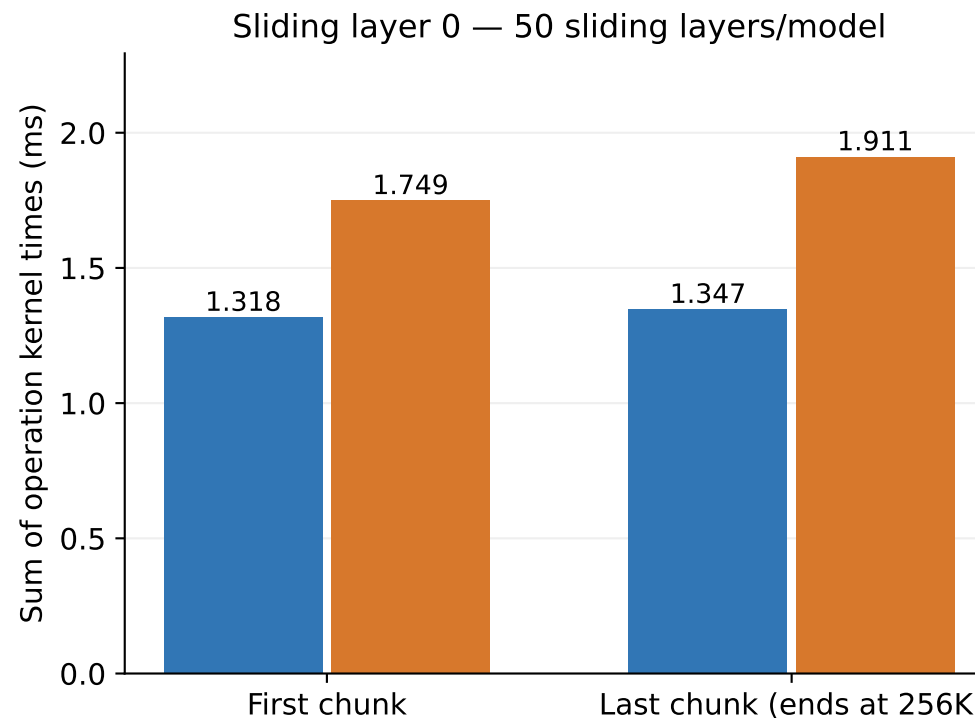
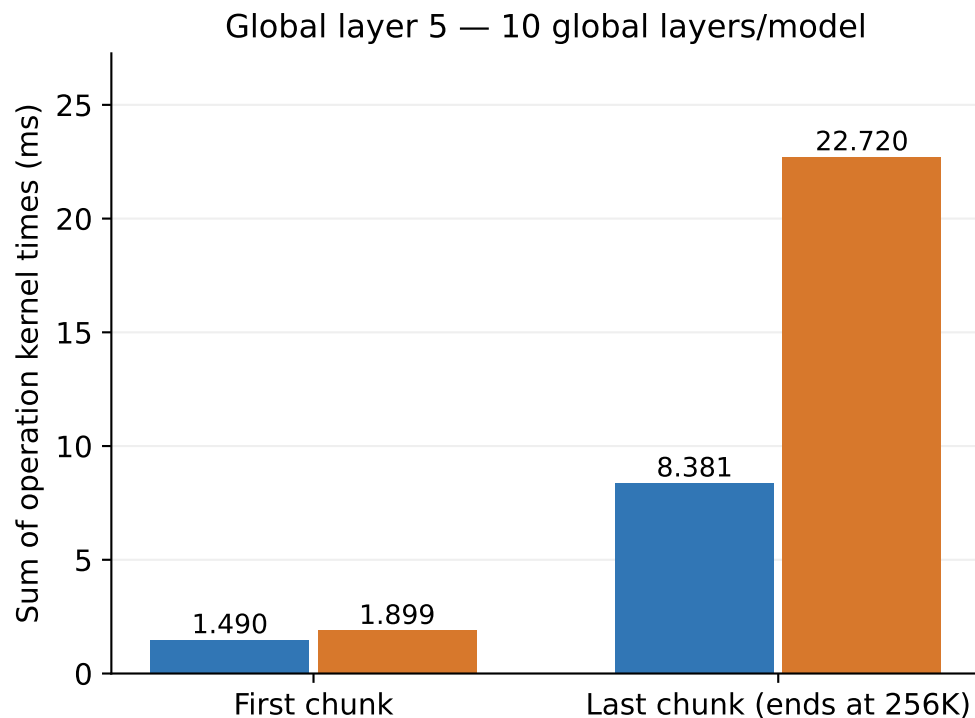
Memory: four 256K request caches fit. KV storage falls from 28.2 to 14.1 GiB/device when reducing eight slots to four. Weights and other buffers are additional. The canonical reference allocates one slot; both use full-history caches.

Validation: 132 host checks and six-layer output/KV, isolation, migration and trace replay checks passed. The strict 60-layer numerical comparison showed drift; further rounding analysis was stopped. Original math settings retained.

Where does batching add time?

Isolated layer profiles. Both paths process 4,096 tokens; lower bars are better.

Canonical 1×4K Batch 4×1K



Global, last: batching adds 14.339 ms per layer. The four attention calls total 21.331 ms versus 7.092 ms for canonical's one call, explaining 99.3% of the increase.

Sliding, last: batching adds 0.564 ms per layer. The four attention calls total 0.693 ms versus 0.231 ms for canonical's one call, explaining 81.7% of the increase.

Prefix control: batched global time is 22.752 ms at a 252K prefix and 22.720 ms at 255K. The extra 3K prefix does not explain the slowdown.

Token ranges ($K = 1024$): first = canonical $[0, 4K)$, batch $4 \times [0, 1K)$. Last = canonical $[252K, 256K)$, batch $4 \times [255K, 256K)$.

These are profiled kernel sums, not full-model latency. The layer test uses random KV histories and token embeddings; the full-model charts use actual model histories. Each panel has its own vertical scale.

Global layer: operation comparison

All times are microseconds, summed over the calls in one layer. C = canonical 1×4K; B = batch 4×1K.

Operation	Calls C	Calls B	First C μ s	First B μ s	Extra μ s	Last C μ s	Last B μ s	Extra μ s
Norm input redistribution	4	4	15.1	15.3	+0.2	15.0	15.9	+0.9
RMS normalization	6	6	64.6	64.4	-0.2	64.7	64.6	-0.1
Norm output redistribution	4	4	16.3	16.1	-0.2	16.2	16.0	-0.2
TP all-gather	2	2	123.4	123.2	-0.2	123.4	124.3	+1.0
Q/K/V projection	1	1	125.3	121.6	-3.7	125.2	121.0	-4.1
Split Q/K/V heads	1	1	36.2	36.5	+0.3	36.2	36.7	+0.5
Local tensor slices	3	11	10.3	54.0	+43.7	10.3	58.5	+48.2
RoPE rotation	1	1	17.4	17.4	+0.0	17.3	17.4	+0.1
Local tensor concatenation	2	3	19.9	45.7	+25.8	19.4	45.3	+25.9
Rotary weight tiling	1	1	5.6	5.6	+0.0	5.6	5.6	+0.0
Elementwise operations	4	4	32.6	32.7	+0.1	32.6	32.9	+0.3
Rotary transform	1	1	7.7	7.7	-0.0	7.5	7.6	+0.1
Rotary-channel gather	1	1	31.5	31.5	+0.0	31.5	31.6	+0.1
KV dtype conversion	1	4	2.5	7.6	+5.1	2.4	8.1	+5.7
KV-cache writes	1	4	7.8	40.8	+33.0	7.8	41.7	+33.9
Attention (SDPA)	1	4	244.4	547.3	+302.9	7,091.9	21,331.2	+14,239.3
Combine attention heads	1	1	36.1	36.4	+0.4	35.5	35.7	+0.3
Attention output projection	1	1	113.1	113.1	+0.0	112.9	113.6	+0.7
TP reduce-scatter	2	2	158.3	158.5	+0.2	201.3	189.3	-12.0
MLP gate projection	1	1	158.4	159.1	+0.7	159.6	158.7	-0.9
MLP up projection	1	1	124.2	124.5	+0.3	125.3	124.6	-0.7
MLP down projection	1	1	139.6	140.0	+0.4	139.4	139.9	+0.5
TOTAL	41	59	1,490.3	1,899.1	+408.8	8,381.1	22,720.2	+14,339.1

The five projection/MLP matmuls have the same packed row count in both paths. Attention, cache writes and local slicing operate per request in the batch. Counts include all four requests.

Slices/concatenation copy rows locally; norm redistribution changes local memory layout. The SDPA operation includes its internal ring KV exchange. The separate TP all-gather/reduce-scatter operations communicate between tensor-parallel devices.

Timing source: tt-perf-report main (version 1.4.1), with the final warmed replay selected by signposts. Ordinary ops use the slowest device; collectives use the device average. Full per-call tables follow.

Sliding-window layer: operation comparison

All times are microseconds, summed over the calls in one layer. C = canonical 1×4K; B = batch 4×1K.

Operation	Calls C	Calls B	First C μ s	First B μ s	Extra μ s	Last C μ s	Last B μ s	Extra μ s
Norm input redistribution	4	4	15.0	15.2	+0.2	15.2	15.2	+0.0
RMS normalization	7	7	62.8	63.1	+0.3	62.4	62.9	+0.5
Norm output redistribution	4	4	16.1	16.2	+0.0	16.0	16.1	+0.1
TP all-gather	2	2	122.1	122.5	+0.4	123.5	122.4	-1.0
Q/K/V projection	1	1	112.6	113.3	+0.7	112.1	112.8	+0.7
Split Q/K/V heads	1	1	29.7	29.7	-0.0	30.0	30.0	+0.0
RoPE rotation	2	2	42.4	42.4	-0.0	42.3	42.5	+0.2
KV dtype conversion	2	8	6.3	21.6	+15.4	6.2	20.5	+14.2
KV-cache writes	2	8	10.1	42.6	+32.5	10.0	42.7	+32.8
Attention (SDPA)	1	4	204.3	530.6	+326.3	231.4	692.6	+461.2
Combine attention heads	1	1	18.7	18.5	-0.1	18.2	18.5	+0.3
Attention output projection	1	1	70.9	70.8	-0.0	71.3	72.6	+1.3
TP reduce-scatter	2	2	154.6	157.3	+2.8	156.8	156.4	-0.4
Elementwise operations	3	3	29.6	29.4	-0.2	29.5	29.7	+0.1
MLP gate projection	1	1	158.5	159.1	+0.6	158.7	160.3	+1.6
MLP up projection	1	1	124.2	124.1	-0.1	123.7	124.3	+0.6
MLP down projection	1	1	139.8	139.8	-0.1	139.7	139.6	-0.1
Local tensor slices	0	12	0.0	35.1	+35.1	0.0	34.4	+34.4
Local tensor concatenation	0	1	0.0	18.0	+18.0	0.0	17.8	+17.8
TOTAL	36	64	1,317.7	1,749.5	+431.7	1,347.0	1,911.3	+564.3

The five projection/MLP matmuls have the same packed row count in both paths. Attention, cache writes and local slicing operate per request in the batch. Counts include all four requests.

Slices/concatenation copy rows locally; norm redistribution changes local memory layout. The SDPA operation includes its internal ring KV exchange. The separate TP all-gather/reduce-scatter operations communicate between tensor-parallel devices.

Timing source: tt-perf-report main (version 1.4.1), with the final warmed replay selected by signposts. Ordinary ops use the slowest device; collectives use the device average. Full per-call tables follow.

Global, first: canonical 1×4K

tt-perf-report per-call table | request tokens [0, 4,096) | kernel sum 1.490 ms

#	Operation / matmul M × K × N	Device μs	Gap μs	Cores	DRAM GB/s	Compute TFLOP/s	Bound	Math	Input 0 storage
1	Interleaved → sharded	4.0	—	24	—	—	—	—	DRAM
2	LayerNorm	8.9	2.8	24	—	—	—	—	L1 block
3	Sharded → interleaved	5.3	1.1	24	—	—	—	—	L1 block
4	AllGather	61.4	4.6	12	—	—	—	—	DRAM
5	Matmul 512 x 5376 x 4608	125.3	0.5	96	279.3	202.5	SLOW	LoFi	DRAM
6	NlpCreateHeads	36.2	0.6	16	—	—	—	—	DRAM
7	LayerNorm	18.9	0.5	120	—	—	—	—	L1
8	LayerNorm	7.6	0.6	16	—	—	—	—	L1
9	Slice	3.2	0.5	120	—	—	—	—	L1
10	Slice	6.0	0.5	120	—	—	—	—	L1
11	RotaryEmbeddingLlama	17.4	0.5	120	—	—	—	—	L1
12	Concat	17.6	0.5	120	—	—	—	—	L1
13	Slice	1.1	0.6	64	—	—	—	—	L1
14	TilizeWithValPadding	5.6	1.3	1	—	—	—	—	DRAM
15	BinaryNg	3.1	0.5	120	—	—	—	—	L1
16	RotaryEmbedding	7.7	0.6	16	—	—	—	—	L1
17	GatherCodegen	31.5	0.5	64	—	—	—	—	L1
18	Concat	2.4	0.5	120	—	—	—	—	L1
19	Typecast	2.5	0.5	120	—	—	—	—	L1
20	UpdatePaddedKvCache	7.8	0.5	16	—	—	—	—	DRAM_ND_SHARDED
21	RingJointSDPA	244.4	4.7	114	—	—	—	—	DRAM
22	NLPConcatHeads	36.1	0.7	16	—	—	—	—	DRAM
23	Matmul 512 x 4096 x 5376	113.1	0.5	96	280.5	199.4	SLOW	LoFi	DRAM
24	ReduceScatter	78.5	0.6	12	—	—	—	—	DRAM
25	Interleaved → sharded	4.0	0.5	24	—	—	—	—	DRAM
26	LayerNorm	9.8	0.5	24	—	—	—	—	L1 block
27	Sharded → interleaved	2.8	0.9	24	—	—	—	—	L1 block
28	BinaryNg	6.3	0.6	120	—	—	—	—	DRAM
29	Interleaved → sharded	3.1	0.6	24	—	—	—	—	L1
30	LayerNorm	9.7	0.6	24	—	—	—	—	L1 block
31	Sharded → interleaved	5.3	1.1	24	—	—	—	—	L1 block
32	AllGather	62.0	0.6	12	—	—	—	—	DRAM
33	Matmul 512 x 5376 x 5376	158.4	0.5	96	217.2	186.9	SLOW	LoFi	DRAM
34	Matmul 512 x 5376 x 5376	124.2	0.6	96	277.0	238.3	SLOW	LoFi	DRAM
35	BinaryNg	16.1	0.5	120	—	—	—	—	L1
36	Matmul 512 x 5376 x 5376	139.6	0.5	96	246.5	212.1	SLOW	LoFi	L1
37	ReduceScatter	79.9	0.6	12	—	—	—	—	DRAM
38	Interleaved → sharded	4.1	0.5	24	—	—	—	—	DRAM
39	LayerNorm	9.7	0.6	24	—	—	—	—	L1 block
40	Sharded → interleaved	2.8	0.9	24	—	—	—	—	L1 block
41	BinaryNg	7.0	0.6	120	—	—	—	—	L1

Source: tt-perf-report 1.4.1, main commit cb9407747a28. Rows follow execution order. The initial idle gap is excluded. Device columns summarize the 32 devices; they are not added together.

DRAM bandwidth, compute throughput and Bound are the tool's estimates. Blank entries mean the tool has no model for that operation; they do not mean zero cost. CSV/text reports retain the complete tool output and advice.

Global, first: batch 4×1K

tt-perf-report per-call table | request tokens [0, 1,024) | kernel sum 1.899 ms

#	Operation / matmul M × K × N	Device µs	Gap µs	Cores	DRAM GB/s	Compute TFLOP/s	Bound	Math	Input 0 storage
1	Interleaved → sharded	4.0	—	24	—	—	—	—	DRAM
2	LayerNorm	8.9	2.8	24	—	—	—	—	L1 block
3	Sharded → interleaved	5.2	1.1	24	—	—	—	—	L1 block
4	AllGather	61.1	4.7	12	—	—	—	—	DRAM
5	Matmul 512 x 5376 x 4608	121.6	0.6	96	287.8	208.6	SLOW	LoFi	DRAM
6	NlpCreateHeads	36.5	0.7	16	—	—	—	—	DRAM
7	LayerNorm	18.9	0.5	120	—	—	—	—	L1
8	LayerNorm	7.5	0.5	16	—	—	—	—	L1
9	Slice	3.2	0.5	120	—	—	—	—	L1
10	Slice	6.1	0.5	120	—	—	—	—	L1
11	RotaryEmbeddingLlama	17.4	0.5	120	—	—	—	—	L1
12	Concat	17.5	0.5	120	—	—	—	—	L1
13	Slice	1.1	0.6	64	—	—	—	—	L1
14	TilizeWithValPadding	5.6	1.2	1	—	—	—	—	DRAM
15	BinaryNg	3.1	0.5	120	—	—	—	—	L1
16	RotaryEmbedding	7.7	0.5	16	—	—	—	—	L1
17	GatherCodegen	31.5	0.5	64	—	—	—	—	L1
18	Concat	2.4	0.5	120	—	—	—	—	L1
19	Slice	10.0	0.5	120	—	—	—	—	DRAM
20	Slice	2.1	0.6	80	—	—	—	—	L1
21	Typecast	2.0	0.7	80	—	—	—	—	DRAM
22	UpdatePaddedKvCache	11.5	1.0	4	—	—	—	—	DRAM_ND_SHARDED
23	RingJointSDPA	135.1	0.6	114	—	—	—	—	DRAM
24	Slice	9.1	0.6	120	—	—	—	—	DRAM
25	Slice	1.5	0.6	80	—	—	—	—	L1
26	Typecast	1.8	0.6	80	—	—	—	—	DRAM
27	UpdatePaddedKvCache	9.9	0.8	4	—	—	—	—	DRAM_ND_SHARDED
28	RingJointSDPA	136.6	0.6	114	—	—	—	—	DRAM
29	Slice	8.9	0.6	120	—	—	—	—	DRAM
30	Slice	1.6	0.6	80	—	—	—	—	L1
31	Typecast	1.8	0.6	80	—	—	—	—	DRAM
32	UpdatePaddedKvCache	9.6	0.9	4	—	—	—	—	DRAM_ND_SHARDED
33	RingJointSDPA	138.2	0.5	114	—	—	—	—	DRAM
34	Slice	8.9	0.6	120	—	—	—	—	DRAM
35	Slice	1.5	0.6	80	—	—	—	—	L1
36	Typecast	2.0	0.7	80	—	—	—	—	DRAM
37	UpdatePaddedKvCache	9.8	0.9	4	—	—	—	—	DRAM_ND_SHARDED
38	RingJointSDPA	137.4	0.5	114	—	—	—	—	DRAM
39	Concat	25.8	0.6	120	—	—	—	—	DRAM
40	NLPConcatHeads	36.4	0.7	16	—	—	—	—	DRAM
41	Matmul 512 x 4096 x 5376	113.1	0.5	96	280.4	199.3	SLOW	LoFi	DRAM
42	ReduceScatter	81.2	0.6	12	—	—	—	—	DRAM
43	Interleaved → sharded	4.1	0.6	24	—	—	—	—	DRAM
44	LayerNorm	9.7	0.5	24	—	—	—	—	L1 block
45	Sharded → interleaved	2.8	1.1	24	—	—	—	—	L1 block
46	BinaryNg	6.4	0.6	120	—	—	—	—	DRAM
47	Interleaved → sharded	3.1	0.5	24	—	—	—	—	L1
48	LayerNorm	9.7	0.6	24	—	—	—	—	L1 block
49	Sharded → interleaved	5.3	0.9	24	—	—	—	—	L1 block
50	AllGather	62.1	0.6	12	—	—	—	—	DRAM
51	Matmul 512 x 5376 x 5376	159.1	0.5	96	216.2	186.0	SLOW	LoFi	DRAM
52	Matmul 512 x 5376 x 5376	124.5	0.6	96	276.4	237.7	SLOW	LoFi	DRAM
53	BinaryNg	16.2	0.6	120	—	—	—	—	L1
54	Matmul 512 x 5376 x 5376	140.0	0.5	96	245.8	211.4	SLOW	LoFi	L1
55	ReduceScatter	77.3	0.6	12	—	—	—	—	DRAM
56	Interleaved → sharded	4.2	0.6	24	—	—	—	—	DRAM
57	LayerNorm	9.7	0.6	24	—	—	—	—	L1 block
58	Sharded → interleaved	2.8	0.9	24	—	—	—	—	L1 block
59	BinaryNg	7.1	0.6	120	—	—	—	—	L1

Source: tt-perf-report 1.4.1, main commit cb9407747a28. Rows follow execution order. The initial idle gap is excluded. Device columns summarize the 32 devices; they are not added together.

DRAM bandwidth, compute throughput and Bound are the tool's estimates. Blank entries mean the tool has no model for that operation; they do not mean zero cost. CSV/text reports retain the complete tool output and advice.

Global, last: canonical 1×4K

tt-perf-report per-call table | request tokens [258,048, 262,144) | kernel sum 8.381 ms

#	Operation / matmul M × K × N	Device μs	Gap μs	Cores	DRAM GB/s	Compute TFLOP/s	Bound	Math	Input 0 storage
1	Interleaved → sharded	4.0	—	24	—	—	—	—	DRAM
2	LayerNorm	8.9	2.9	24	—	—	—	—	L1 block
3	Sharded → interleaved	5.2	1.1	24	—	—	—	—	L1 block
4	AllGather	61.0	4.5	12	—	—	—	—	DRAM
5	Matmul 512 x 5376 x 4608	125.2	0.5	96	279.6	202.6	SLOW	LoFi	DRAM
6	NlpCreateHeads	36.2	0.6	16	—	—	—	—	DRAM
7	LayerNorm	18.9	0.5	120	—	—	—	—	L1
8	LayerNorm	7.6	0.5	16	—	—	—	—	L1
9	Slice	3.2	0.6	120	—	—	—	—	L1
10	Slice	6.0	0.5	120	—	—	—	—	L1
11	RotaryEmbeddingLlama	17.3	0.5	120	—	—	—	—	L1
12	Concat	17.1	0.5	120	—	—	—	—	L1
13	Slice	1.1	0.6	64	—	—	—	—	L1
14	TilizeWithValPadding	5.6	1.3	1	—	—	—	—	DRAM
15	BinaryNg	3.2	0.5	120	—	—	—	—	L1
16	RotaryEmbedding	7.5	0.6	16	—	—	—	—	L1
17	GatherCodegen	31.5	0.5	64	—	—	—	—	L1
18	Concat	2.4	0.5	120	—	—	—	—	L1
19	Typecast	2.4	0.5	120	—	—	—	—	L1
20	UpdatePaddedKvCache	7.8	0.5	16	—	—	—	—	DRAM_ND_SHARDED
21	RingJointSDPA	7,091.9	4.8	114	—	—	—	—	DRAM
22	NLPConcatHeads	35.5	0.6	16	—	—	—	—	DRAM
23	Matmul 512 x 4096 x 5376	112.9	0.5	96	281.0	199.8	SLOW	LoFi	DRAM
24	ReduceScatter	122.2	0.6	12	—	—	—	—	DRAM
25	Interleaved → sharded	4.0	0.5	24	—	—	—	—	DRAM
26	LayerNorm	9.9	0.6	24	—	—	—	—	L1 block
27	Sharded → interleaved	2.8	0.9	24	—	—	—	—	L1 block
28	BinaryNg	6.4	0.7	120	—	—	—	—	DRAM
29	Interleaved → sharded	3.1	0.5	24	—	—	—	—	L1
30	LayerNorm	9.7	0.5	24	—	—	—	—	L1 block
31	Sharded → interleaved	5.4	1.1	24	—	—	—	—	L1 block
32	AllGather	62.4	0.6	12	—	—	—	—	DRAM
33	Matmul 512 x 5376 x 5376	159.6	0.5	96	215.6	185.4	SLOW	LoFi	DRAM
34	Matmul 512 x 5376 x 5376	125.3	0.6	96	274.5	236.1	SLOW	LoFi	DRAM
35	BinaryNg	16.2	0.6	120	—	—	—	—	L1
36	Matmul 512 x 5376 x 5376	139.4	0.5	96	246.8	212.3	SLOW	LoFi	L1
37	ReduceScatter	79.1	0.6	12	—	—	—	—	DRAM
38	Interleaved → sharded	4.0	0.6	24	—	—	—	—	DRAM
39	LayerNorm	9.7	0.6	24	—	—	—	—	L1 block
40	Sharded → interleaved	2.7	1.0	24	—	—	—	—	L1 block
41	BinaryNg	6.9	0.6	120	—	—	—	—	L1

Source: tt-perf-report 1.4.1, main commit cb9407747a28. Rows follow execution order. The initial idle gap is excluded. Device columns summarize the 32 devices; they are not added together.

DRAM bandwidth, compute throughput and Bound are the tool's estimates. Blank entries mean the tool has no model for that operation; they do not mean zero cost. CSV/text reports retain the complete tool output and advice.

Global, last: batch 4×1K

tt-perf-report per-call table | request tokens [261,120, 262,144) | kernel sum 22.720 ms

#	Operation / matmul M × K × N	Device µs	Gap µs	Cores	DRAM GB/s	Compute TFLOP/s	Bound	Math	Input 0 storage
1	Interleaved → sharded	4.3	—	24	—	—	—	—	DRAM
2	LayerNorm	8.9	2.7	24	—	—	—	—	L1 block
3	Sharded → interleaved	5.2	1.2	24	—	—	—	—	L1 block
4	AllGather	61.3	4.6	12	—	—	—	—	DRAM
5	Matmul 512 x 5376 x 4608	121.0	0.5	96	289.1	209.6	SLOW	LoFi	DRAM
6	NlpCreateHeads	36.7	0.6	16	—	—	—	—	DRAM
7	LayerNorm	18.9	0.5	120	—	—	—	—	L1
8	LayerNorm	7.5	0.5	16	—	—	—	—	L1
9	Slice	3.2	0.5	120	—	—	—	—	L1
10	Slice	6.0	0.5	120	—	—	—	—	L1
11	RotaryEmbeddingLlama	17.4	0.5	120	—	—	—	—	L1
12	Concat	17.3	0.5	120	—	—	—	—	L1
13	Slice	1.1	0.6	64	—	—	—	—	L1
14	TilizeWithValPadding	5.6	1.2	1	—	—	—	—	DRAM
15	BinaryNg	3.1	0.5	120	—	—	—	—	L1
16	RotaryEmbedding	7.6	0.5	16	—	—	—	—	L1
17	GatherCodegen	31.6	0.6	64	—	—	—	—	L1
18	Concat	2.4	0.5	120	—	—	—	—	L1
19	Slice	10.4	0.5	120	—	—	—	—	DRAM
20	Slice	2.1	0.5	80	—	—	—	—	L1
21	Typecast	2.0	0.5	80	—	—	—	—	DRAM
22	UpdatePaddedKvCache	11.3	0.6	4	—	—	—	—	DRAM_ND_SHARDED
23	RingJointSDPA	5,315.0	0.6	114	—	—	—	—	DRAM
24	Slice	10.5	0.6	120	—	—	—	—	DRAM
25	Slice	1.7	0.5	80	—	—	—	—	L1
26	Typecast	2.1	0.7	80	—	—	—	—	DRAM
27	UpdatePaddedKvCache	10.2	0.8	4	—	—	—	—	DRAM_ND_SHARDED
28	RingJointSDPA	5,324.5	0.6	114	—	—	—	—	DRAM
29	Slice	10.5	0.6	120	—	—	—	—	DRAM
30	Slice	1.5	0.6	80	—	—	—	—	L1
31	Typecast	2.0	0.8	80	—	—	—	—	DRAM
32	UpdatePaddedKvCache	10.1	0.8	4	—	—	—	—	DRAM_ND_SHARDED
33	RingJointSDPA	5,338.5	0.6	114	—	—	—	—	DRAM
34	Slice	10.0	0.6	120	—	—	—	—	DRAM
35	Slice	1.6	0.6	80	—	—	—	—	L1
36	Typecast	1.9	0.6	80	—	—	—	—	DRAM
37	UpdatePaddedKvCache	10.1	0.8	4	—	—	—	—	DRAM_ND_SHARDED
38	RingJointSDPA	5,353.2	0.6	114	—	—	—	—	DRAM
39	Concat	25.7	0.6	120	—	—	—	—	DRAM
40	NLPConcatHeads	35.7	0.7	16	—	—	—	—	DRAM
41	Matmul 512 x 4096 x 5376	113.6	0.5	96	279.3	198.6	SLOW	LoFi	DRAM
42	ReduceScatter	110.8	0.6	12	—	—	—	—	DRAM
43	Interleaved → sharded	4.1	0.6	24	—	—	—	—	DRAM
44	LayerNorm	9.9	0.6	24	—	—	—	—	L1 block
45	Sharded → interleaved	2.8	1.1	24	—	—	—	—	L1 block
46	BinaryNg	6.5	0.6	120	—	—	—	—	DRAM
47	Interleaved → sharded	3.1	0.5	24	—	—	—	—	L1
48	LayerNorm	9.7	0.5	24	—	—	—	—	L1 block
49	Sharded → interleaved	5.2	1.1	24	—	—	—	—	L1 block
50	AllGather	63.0	0.6	12	—	—	—	—	DRAM
51	Matmul 512 x 5376 x 5376	158.7	0.5	96	216.8	186.5	SLOW	LoFi	DRAM
52	Matmul 512 x 5376 x 5376	124.6	0.6	96	276.1	237.5	SLOW	LoFi	DRAM
53	BinaryNg	16.2	0.6	120	—	—	—	—	L1
54	Matmul 512 x 5376 x 5376	139.9	0.5	96	245.9	211.5	SLOW	LoFi	L1
55	ReduceScatter	78.5	0.6	12	—	—	—	—	DRAM
56	Interleaved → sharded	4.3	0.5	24	—	—	—	—	DRAM
57	LayerNorm	9.7	0.6	24	—	—	—	—	L1 block
58	Sharded → interleaved	2.7	0.9	24	—	—	—	—	L1 block
59	BinaryNg	7.1	0.7	120	—	—	—	—	L1

Source: tt-perf-report 1.4.1, main commit cb9407747a28. Rows follow execution order. The initial idle gap is excluded. Device columns summarize the 32 devices; they are not added together.

DRAM bandwidth, compute throughput and Bound are the tool's estimates. Blank entries mean the tool has no model for that operation; they do not mean zero cost. CSV/text reports retain the complete tool output and advice.

Sliding, first: canonical 1×4K

tt-perf-report per-call table | request tokens [0, 4,096) | kernel sum 1.318 ms

#	Operation / matmul M × K × N	Device μs	Gap μs	Cores	DRAM GB/s	Compute TFLOP/s	Bound	Math	Input 0 storage
1	Interleaved → sharded	4.0	—	24	—	—	—	—	DRAM
2	LayerNorm	8.9	2.9	24	—	—	—	—	L1 block
3	Sharded → interleaved	5.3	1.1	24	—	—	—	—	L1 block
4	AllGather	61.1	4.7	12	—	—	—	—	DRAM
5	Matmul 512 x 5376 x 4096	112.6	0.6	96	281.7	200.3	SLOW	LoFi	DRAM
6	NlpCreateHeads	29.7	0.6	16	—	—	—	—	DRAM
7	LayerNorm	11.7	0.5	120	—	—	—	—	L1
8	LayerNorm	6.8	0.5	64	—	—	—	—	L1
9	LayerNorm	6.3	0.5	64	—	—	—	—	L1
10	RotaryEmbeddingLlama	25.4	0.5	120	—	—	—	—	L1
11	RotaryEmbeddingLlama	17.0	0.5	120	—	—	—	—	L1
12	Typecast	3.2	0.5	120	—	—	—	—	L1
13	UpdatePaddedKvCache	5.1	0.5	64	—	—	—	—	DRAM_ND_SHARDED
14	Typecast	3.1	0.6	120	—	—	—	—	L1
15	UpdatePaddedKvCache	5.1	0.5	64	—	—	—	—	DRAM_ND_SHARDED
16	RingJointSDPA	204.3	9.6	112	—	—	—	—	DRAM
17	NLPConcatHeads	18.7	0.6	16	—	—	—	—	DRAM
18	Matmul 512 x 2048 x 5376	70.9	0.5	96	262.6	159.1	SLOW	LoFi	DRAM
19	ReduceScatter	76.5	0.6	12	—	—	—	—	DRAM
20	Interleaved → sharded	4.0	0.5	24	—	—	—	—	DRAM
21	LayerNorm	9.7	0.6	24	—	—	—	—	L1 block
22	Sharded → interleaved	2.9	1.1	24	—	—	—	—	L1 block
23	BinaryNg	6.4	0.6	120	—	—	—	—	DRAM
24	Interleaved → sharded	3.1	0.5	24	—	—	—	—	L1
25	LayerNorm	9.7	0.6	24	—	—	—	—	L1 block
26	Sharded → interleaved	5.3	1.0	24	—	—	—	—	L1 block
27	AllGather	61.0	0.7	12	—	—	—	—	DRAM
28	Matmul 512 x 5376 x 5376	158.5	0.6	96	217.1	186.7	SLOW	LoFi	DRAM
29	Matmul 512 x 5376 x 5376	124.2	0.6	96	277.0	238.3	SLOW	LoFi	DRAM
30	BinaryNg	16.2	0.6	120	—	—	—	—	L1
31	Matmul 512 x 5376 x 5376	139.8	0.6	96	246.0	211.6	SLOW	LoFi	L1
32	ReduceScatter	78.1	0.6	12	—	—	—	—	DRAM
33	Interleaved → sharded	3.9	0.5	24	—	—	—	—	DRAM
34	LayerNorm	9.7	0.6	24	—	—	—	—	L1 block
35	Sharded → interleaved	2.8	1.1	24	—	—	—	—	L1 block
36	BinaryNg	7.0	0.7	120	—	—	—	—	L1

Source: tt-perf-report 1.4.1, main commit cb9407747a28. Rows follow execution order. The initial idle gap is excluded. Device columns summarize the 32 devices; they are not added together.

DRAM bandwidth, compute throughput and Bound are the tool's estimates. Blank entries mean the tool has no model for that operation; they do not mean zero cost. CSV/text reports retain the complete tool output and advice.

Sliding, first: batch 4×1K

tt-perf-report per-call table | request tokens [0, 1,024) | kernel sum 1.749 ms

#	Operation / matmul M × K × N	Device μs	Gap μs	Cores	DRAM GB/s	Compute TFLOP/s	Bound	Math	Input 0 storage
1	Interleaved → sharded	4.0	—	24	—	—	—	—	DRAM
2	LayerNorm	8.9	2.8	24	—	—	—	—	L1 block
3	Sharded → interleaved	5.3	1.1	24	—	—	—	—	L1 block
4	AllGather	60.7	4.7	12	—	—	—	—	DRAM
5	Matmul 512 x 5376 x 4096	113.3	0.6	96	280.0	199.1	SLOW	LoFi	DRAM
6	NlpCreateHeads	29.7	0.6	16	—	—	—	—	DRAM
7	LayerNorm	11.9	0.5	120	—	—	—	—	L1
8	LayerNorm	6.8	0.5	64	—	—	—	—	L1
9	LayerNorm	6.2	0.5	64	—	—	—	—	L1
10	RotaryEmbeddingLlama	25.5	0.5	120	—	—	—	—	L1
11	RotaryEmbeddingLlama	16.9	0.5	120	—	—	—	—	L1
12	Slice	4.2	0.6	120	—	—	—	—	DRAM
13	Slice	2.0	0.5	120	—	—	—	—	L1
14	Slice	2.0	0.5	120	—	—	—	—	L1
15	Typecast	2.9	0.5	120	—	—	—	—	DRAM
16	UpdatePaddedKvCache	5.3	0.5	16	—	—	—	—	DRAM_ND_SHARDED
17	Typecast	2.3	0.5	120	—	—	—	—	DRAM
18	UpdatePaddedKvCache	5.4	0.5	16	—	—	—	—	DRAM_ND_SHARDED
19	RingJointSDPA	157.2	0.5	112	—	—	—	—	DRAM
20	Slice	4.4	0.6	120	—	—	—	—	DRAM
21	Slice	2.0	0.5	120	—	—	—	—	L1
22	Slice	2.0	0.5	120	—	—	—	—	L1
23	Typecast	2.6	0.5	120	—	—	—	—	DRAM
24	UpdatePaddedKvCache	5.2	0.5	16	—	—	—	—	DRAM_ND_SHARDED
25	Typecast	2.5	0.6	120	—	—	—	—	DRAM
26	UpdatePaddedKvCache	5.3	0.5	16	—	—	—	—	DRAM_ND_SHARDED
27	RingJointSDPA	140.4	0.5	114	—	—	—	—	DRAM
28	Slice	5.1	0.6	120	—	—	—	—	DRAM
29	Slice	2.0	0.5	120	—	—	—	—	L1
30	Slice	2.9	0.5	120	—	—	—	—	L1
31	Typecast	3.1	0.5	120	—	—	—	—	DRAM
32	UpdatePaddedKvCache	5.3	0.5	16	—	—	—	—	DRAM_ND_SHARDED
33	Typecast	2.6	0.5	120	—	—	—	—	DRAM
34	UpdatePaddedKvCache	5.4	0.5	16	—	—	—	—	DRAM_ND_SHARDED
35	RingJointSDPA	116.5	0.5	114	—	—	—	—	DRAM
36	Slice	4.4	0.6	120	—	—	—	—	DRAM
37	Slice	2.1	0.5	120	—	—	—	—	L1
38	Slice	2.0	0.5	120	—	—	—	—	L1
39	Typecast	2.5	0.5	120	—	—	—	—	DRAM
40	UpdatePaddedKvCache	5.2	0.5	16	—	—	—	—	DRAM_ND_SHARDED
41	Typecast	3.1	0.6	120	—	—	—	—	DRAM
42	UpdatePaddedKvCache	5.3	0.5	16	—	—	—	—	DRAM_ND_SHARDED
43	RingJointSDPA	116.5	0.5	114	—	—	—	—	DRAM
44	Concat	18.0	0.6	120	—	—	—	—	DRAM
45	NLPConcatHeads	18.5	0.7	16	—	—	—	—	DRAM
46	Matmul 512 x 2048 x 5376	70.8	0.5	96	262.8	159.2	SLOW	LoFi	DRAM
47	ReduceScatter	78.6	0.6	12	—	—	—	—	DRAM
48	Interleaved → sharded	4.1	0.6	24	—	—	—	—	DRAM
49	LayerNorm	9.8	0.5	24	—	—	—	—	L1 block
50	Sharded → interleaved	2.8	1.1	24	—	—	—	—	L1 block
51	BinaryNg	6.3	0.6	120	—	—	—	—	DRAM
52	Interleaved → sharded	3.1	0.5	24	—	—	—	—	L1
53	LayerNorm	9.7	0.6	24	—	—	—	—	L1 block
54	Sharded → interleaved	5.4	1.1	24	—	—	—	—	L1 block
55	AllGather	61.9	0.6	12	—	—	—	—	DRAM
56	Matmul 512 x 5376 x 5376	159.1	0.5	96	216.3	186.0	SLOW	LoFi	DRAM
57	Matmul 512 x 5376 x 5376	124.1	0.5	96	277.2	238.5	SLOW	LoFi	DRAM
58	BinaryNg	16.2	0.6	120	—	—	—	—	L1
59	Matmul 512 x 5376 x 5376	139.8	0.6	96	246.2	211.7	SLOW	LoFi	L1
60	ReduceScatter	78.7	0.6	12	—	—	—	—	DRAM
61	Interleaved → sharded	4.1	0.6	24	—	—	—	—	DRAM
62	LayerNorm	9.8	0.6	24	—	—	—	—	L1 block
63	Sharded → interleaved	2.7	1.0	24	—	—	—	—	L1 block
64	BinaryNg	6.9	0.6	120	—	—	—	—	L1

Source: tt-perf-report 1.4.1, main commit cb9407747a28. Rows follow execution order. The initial idle gap is excluded. Device columns summarize the 32 devices; they are not added together.

DRAM bandwidth, compute throughput and Bound are the tool's estimates. Blank entries mean the tool has no model for that operation; they do not mean zero cost. CSV/text reports retain the complete tool output and advice.

Sliding, last: canonical 1×4K

tt-perf-report per-call table | request tokens [258,048, 262,144) | kernel sum 1.347 ms

#	Operation / matmul M × K × N	Device μs	Gap μs	Cores	DRAM GB/s	Compute TFLOP/s	Bound	Math	Input 0 storage
1	Interleaved → sharded	3.9	—	24	—	—	—	—	DRAM
2	LayerNorm	9.0	2.8	24	—	—	—	—	L1 block
3	Sharded → interleaved	5.2	1.1	24	—	—	—	—	L1 block
4	AllGather	60.8	4.7	12	—	—	—	—	DRAM
5	Matmul 512 x 5376 x 4096	112.1	0.5	96	283.0	201.2	SLOW	LoFi	DRAM
6	NlpCreateHeads	30.0	0.6	16	—	—	—	—	DRAM
7	LayerNorm	11.5	0.5	120	—	—	—	—	L1
8	LayerNorm	6.7	0.5	64	—	—	—	—	L1
9	LayerNorm	6.2	0.5	64	—	—	—	—	L1
10	RotaryEmbeddingLlama	25.4	0.5	120	—	—	—	—	L1
11	RotaryEmbeddingLlama	16.9	0.5	120	—	—	—	—	L1
12	Typecast	3.1	0.5	120	—	—	—	—	L1
13	UpdatePaddedKvCache	5.0	0.5	64	—	—	—	—	DRAM_ND_SHARDED
14	Typecast	3.1	0.6	120	—	—	—	—	L1
15	UpdatePaddedKvCache	4.9	0.5	64	—	—	—	—	DRAM_ND_SHARDED
16	RingJointSDPA	231.4	9.5	112	—	—	—	—	DRAM
17	NLPConcatHeads	18.2	0.6	16	—	—	—	—	DRAM
18	Matmul 512 x 2048 x 5376	71.3	0.5	96	261.0	158.1	SLOW	LoFi	DRAM
19	ReduceScatter	77.4	0.6	12	—	—	—	—	DRAM
20	Interleaved → sharded	4.0	0.6	24	—	—	—	—	DRAM
21	LayerNorm	9.8	0.6	24	—	—	—	—	L1 block
22	Sharded → interleaved	2.7	1.1	24	—	—	—	—	L1 block
23	BinaryNg	6.3	0.7	120	—	—	—	—	DRAM
24	Interleaved → sharded	3.1	0.5	24	—	—	—	—	L1
25	LayerNorm	9.7	0.6	24	—	—	—	—	L1 block
26	Sharded → interleaved	5.3	1.1	24	—	—	—	—	L1 block
27	AllGather	62.6	0.7	12	—	—	—	—	DRAM
28	Matmul 512 x 5376 x 5376	158.7	0.5	96	216.8	186.5	SLOW	LoFi	DRAM
29	Matmul 512 x 5376 x 5376	123.7	0.6	96	278.2	239.3	SLOW	LoFi	DRAM
30	BinaryNg	16.3	0.6	120	—	—	—	—	L1
31	Matmul 512 x 5376 x 5376	139.7	0.6	96	246.3	211.8	SLOW	LoFi	L1
32	ReduceScatter	79.4	0.6	12	—	—	—	—	DRAM
33	Interleaved → sharded	4.1	0.5	24	—	—	—	—	DRAM
34	LayerNorm	9.7	0.6	24	—	—	—	—	L1 block
35	Sharded → interleaved	2.8	1.0	24	—	—	—	—	L1 block
36	BinaryNg	7.0	0.6	120	—	—	—	—	L1

Source: tt-perf-report 1.4.1, main commit cb9407747a28. Rows follow execution order. The initial idle gap is excluded. Device columns summarize the 32 devices; they are not added together.

DRAM bandwidth, compute throughput and Bound are the tool's estimates. Blank entries mean the tool has no model for that operation; they do not mean zero cost. CSV/text reports retain the complete tool output and advice.

Sliding, last: batch 4×1K

tt-perf-report per-call table | request tokens [261,120, 262,144) | kernel sum 1.911 ms

#	Operation / matmul M × K × N	Device μs	Gap μs	Cores	DRAM GB/s	Compute TFLOP/s	Bound	Math	Input 0 storage
1	Interleaved → sharded	4.1	—	24	—	—	—	—	DRAM
2	LayerNorm	8.9	2.7	24	—	—	—	—	L1 block
3	Sharded → interleaved	5.3	1.1	24	—	—	—	—	L1 block
4	AllGather	60.7	4.6	12	—	—	—	—	DRAM
5	Matmul 512 x 5376 x 4096	112.8	0.5	96	281.2	199.9	SLOW	LoFi	DRAM
6	NlpCreateHeads	30.0	0.6	16	—	—	—	—	DRAM
7	LayerNorm	11.7	0.5	120	—	—	—	—	L1
8	LayerNorm	6.9	0.6	64	—	—	—	—	L1
9	LayerNorm	6.2	0.5	64	—	—	—	—	L1
10	RotaryEmbeddingLlama	25.5	0.5	120	—	—	—	—	L1
11	RotaryEmbeddingLlama	17.0	0.5	120	—	—	—	—	L1
12	Slice	4.3	0.5	120	—	—	—	—	DRAM
13	Slice	2.0	0.5	120	—	—	—	—	L1
14	Slice	2.0	0.5	120	—	—	—	—	L1
15	Typecast	2.9	0.4	120	—	—	—	—	DRAM
16	UpdatePaddedKvCache	5.4	0.5	16	—	—	—	—	DRAM_ND_SHARDED
17	Typecast	2.5	0.5	120	—	—	—	—	DRAM
18	UpdatePaddedKvCache	5.4	0.5	16	—	—	—	—	DRAM_ND_SHARDED
19	RingJointSDPA	185.3	0.4	112	—	—	—	—	DRAM
20	Slice	4.3	0.6	120	—	—	—	—	DRAM
21	Slice	2.0	0.5	120	—	—	—	—	L1
22	Slice	2.0	0.5	120	—	—	—	—	L1
23	Typecast	2.4	0.5	120	—	—	—	—	DRAM
24	UpdatePaddedKvCache	5.2	0.5	16	—	—	—	—	DRAM_ND_SHARDED
25	Typecast	2.4	0.5	120	—	—	—	—	DRAM
26	UpdatePaddedKvCache	5.4	0.5	16	—	—	—	—	DRAM_ND_SHARDED
27	RingJointSDPA	169.4	0.5	114	—	—	—	—	DRAM
28	Slice	4.6	0.6	120	—	—	—	—	DRAM
29	Slice	2.1	0.5	120	—	—	—	—	L1
30	Slice	3.0	0.6	120	—	—	—	—	L1
31	Typecast	3.0	0.5	120	—	—	—	—	DRAM
32	UpdatePaddedKvCache	5.3	0.5	16	—	—	—	—	DRAM_ND_SHARDED
33	Typecast	2.4	0.5	120	—	—	—	—	DRAM
34	UpdatePaddedKvCache	5.3	0.5	16	—	—	—	—	DRAM_ND_SHARDED
35	RingJointSDPA	169.6	0.5	114	—	—	—	—	DRAM
36	Slice	4.2	0.6	120	—	—	—	—	DRAM
37	Slice	2.1	0.5	120	—	—	—	—	L1
38	Slice	2.0	0.5	120	—	—	—	—	L1
39	Typecast	2.5	0.5	120	—	—	—	—	DRAM
40	UpdatePaddedKvCache	5.2	0.5	16	—	—	—	—	DRAM_ND_SHARDED
41	Typecast	2.4	0.6	120	—	—	—	—	DRAM
42	UpdatePaddedKvCache	5.5	0.5	16	—	—	—	—	DRAM_ND_SHARDED
43	RingJointSDPA	168.3	0.5	114	—	—	—	—	DRAM
44	Concat	17.8	0.6	120	—	—	—	—	DRAM
45	NLPConcatHeads	18.5	0.7	16	—	—	—	—	DRAM
46	Matmul 512 x 2048 x 5376	72.6	0.5	96	256.5	155.4	SLOW	LoFi	DRAM
47	ReduceScatter	77.6	0.6	12	—	—	—	—	DRAM
48	Interleaved → sharded	4.0	0.6	24	—	—	—	—	DRAM
49	LayerNorm	9.8	0.6	24	—	—	—	—	L1 block
50	Sharded → interleaved	2.8	1.0	24	—	—	—	—	L1 block
51	BinaryNg	6.3	0.7	120	—	—	—	—	DRAM
52	Interleaved → sharded	3.1	0.5	24	—	—	—	—	L1
53	LayerNorm	9.7	0.6	24	—	—	—	—	L1 block
54	Sharded → interleaved	5.2	1.1	24	—	—	—	—	L1 block
55	AllGather	61.7	0.6	12	—	—	—	—	DRAM
56	Matmul 512 x 5376 x 5376	160.3	0.6	96	214.6	184.6	SLOW	LoFi	DRAM
57	Matmul 512 x 5376 x 5376	124.3	0.6	96	276.9	238.2	SLOW	LoFi	DRAM
58	BinaryNg	16.2	0.6	120	—	—	—	—	L1
59	Matmul 512 x 5376 x 5376	139.6	0.5	96	246.5	212.0	SLOW	LoFi	L1
60	ReduceScatter	78.8	0.6	12	—	—	—	—	DRAM
61	Interleaved → sharded	4.0	0.6	24	—	—	—	—	DRAM
62	LayerNorm	9.7	0.5	24	—	—	—	—	L1 block
63	Sharded → interleaved	2.8	0.9	24	—	—	—	—	L1 block
64	BinaryNg	7.2	0.6	120	—	—	—	—	L1

Source: tt-perf-report 1.4.1, main commit cb9407747a28. Rows follow execution order. The initial idle gap is excluded. Device columns summarize the 32 devices; they are not added together.

DRAM bandwidth, compute throughput and Bound are the tool's estimates. Blank entries mean the tool has no model for that operation; they do not mean zero cost. CSV/text reports retain the complete tool output and advice.